



Welcome to the Biodiversity Genomics course!

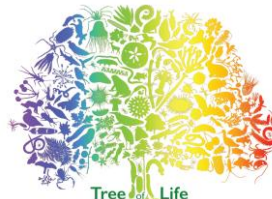
Joana Meier, Karin Näsval, Nicol Rueda, Fernando Seixas
Tree of Life Programme, Wellcome Sanger Institute

Gustavo Silva
Universidad Nacional de Colombia

THE
ROYAL
SOCIETY

The
Branco Weiss
Fellowship
Society in Science

W
wellcome



wellcome
sanger
institute



UNIVERSITY OF
CAMBRIDGE

Course teachers

https://github.com/rapidspeciation/biodiversity_genomics_course



Joana Meier



Nicol Rueda



Karin Näsvall



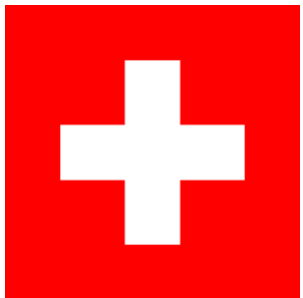
Fernando Seixas



Gustavo Silva

Joana Meier

Childhood



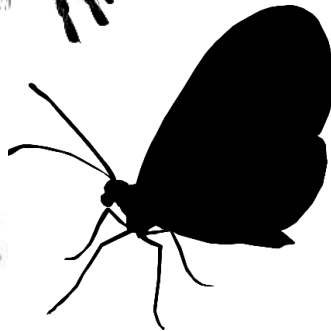
**PhD on cichlid fish
speciation in Bern,
Switzerland**



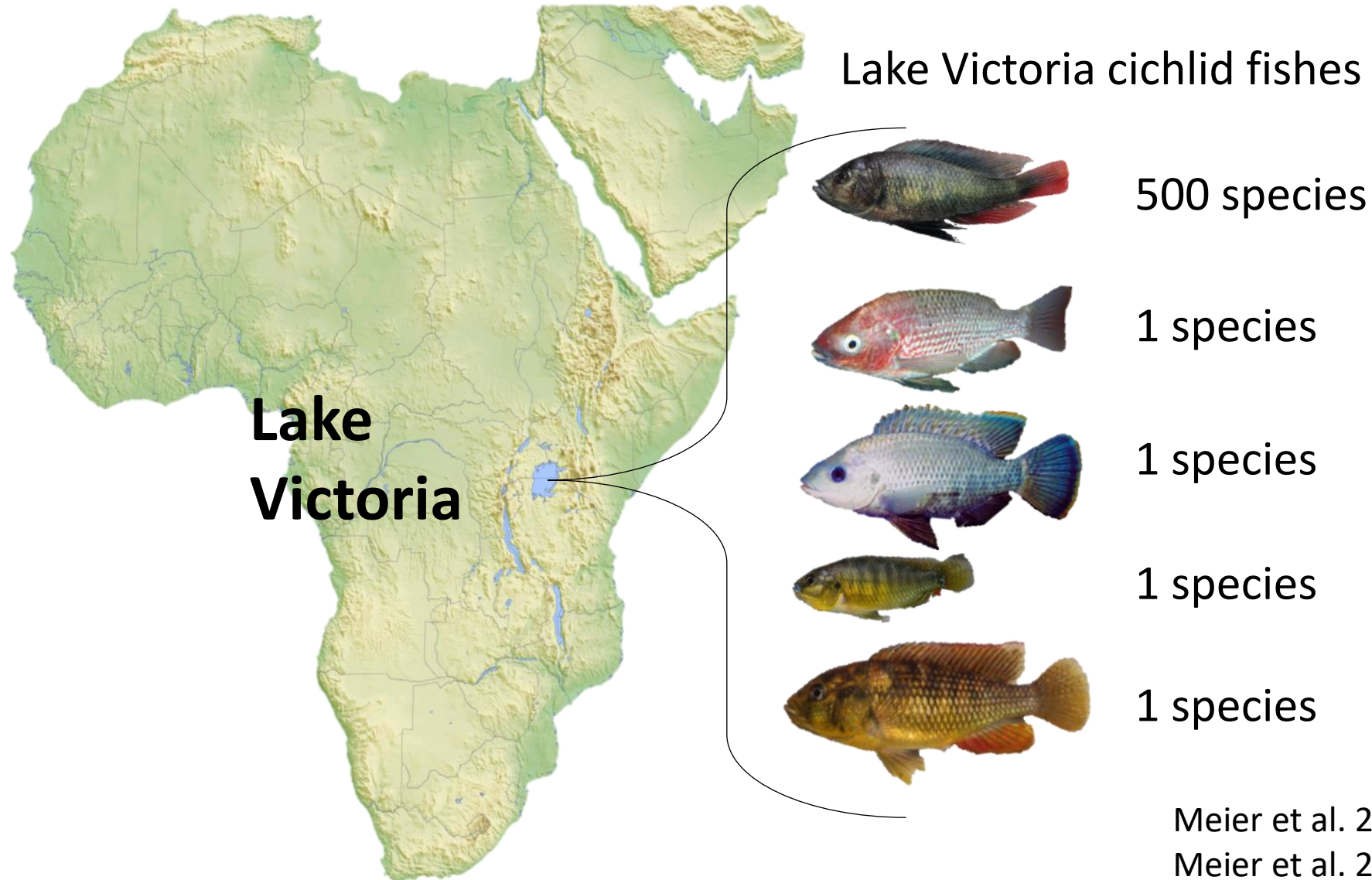
**Postdoc at the
University of
Cambridge, UK**



**Group leader at the
Wellcome Sanger
Institute since 2022**



Why do some lineages diversify very fast?



Meier et al. 2017, Nature Communications
Meier et al. 2023, Science

Nicol Rueda

Postdoctoral fellow

Rapid speciation group, Tree of life

Wellcome Sanger Institute

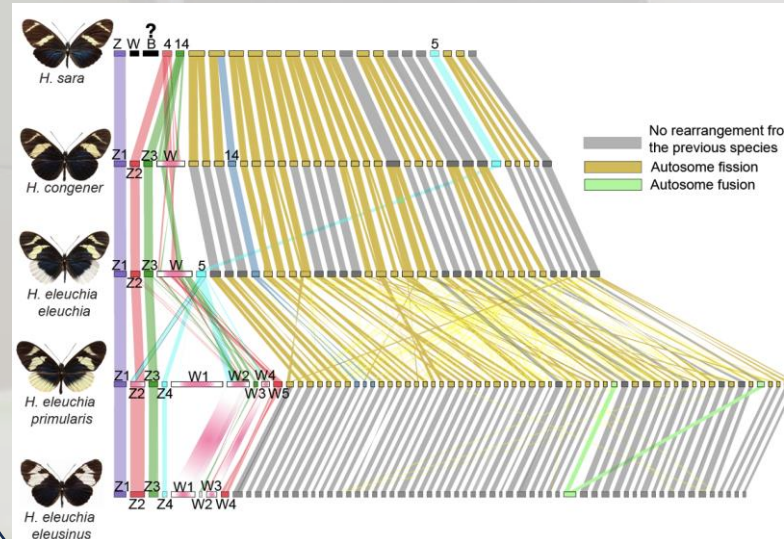
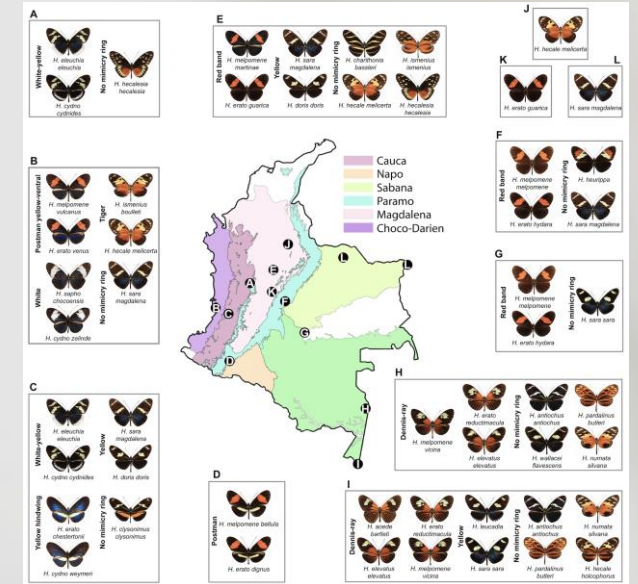
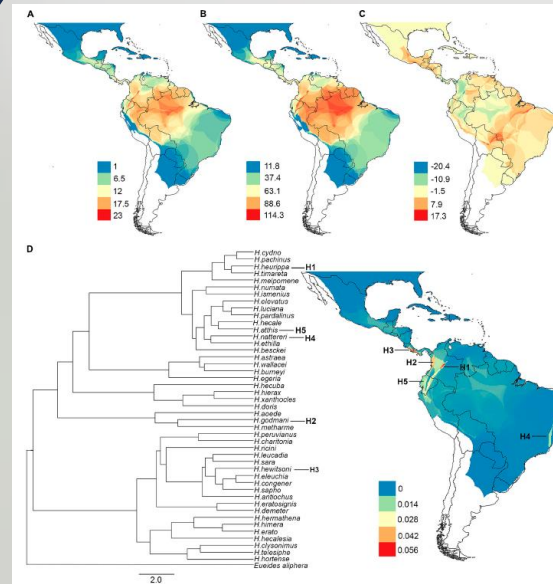
Master of science in Biology in Col/Brazil



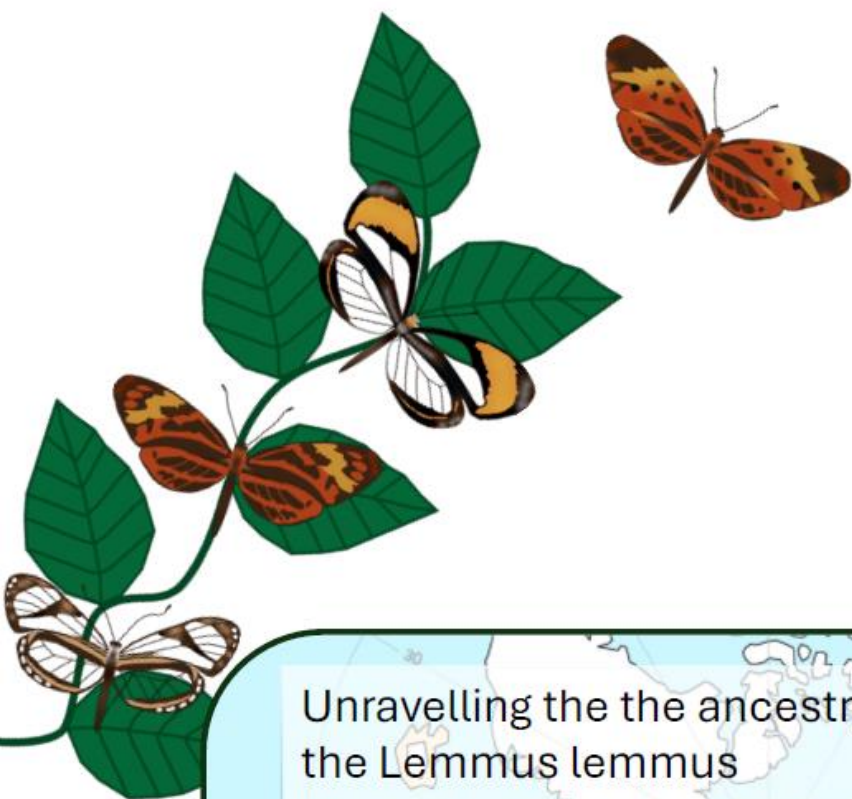
Population monitoring for 7 months in two forests of different conservation status.

Capture-Mark-recapture technique

PhD in Biology in Colombia



Sex chromosome evolution in Lepidoptera



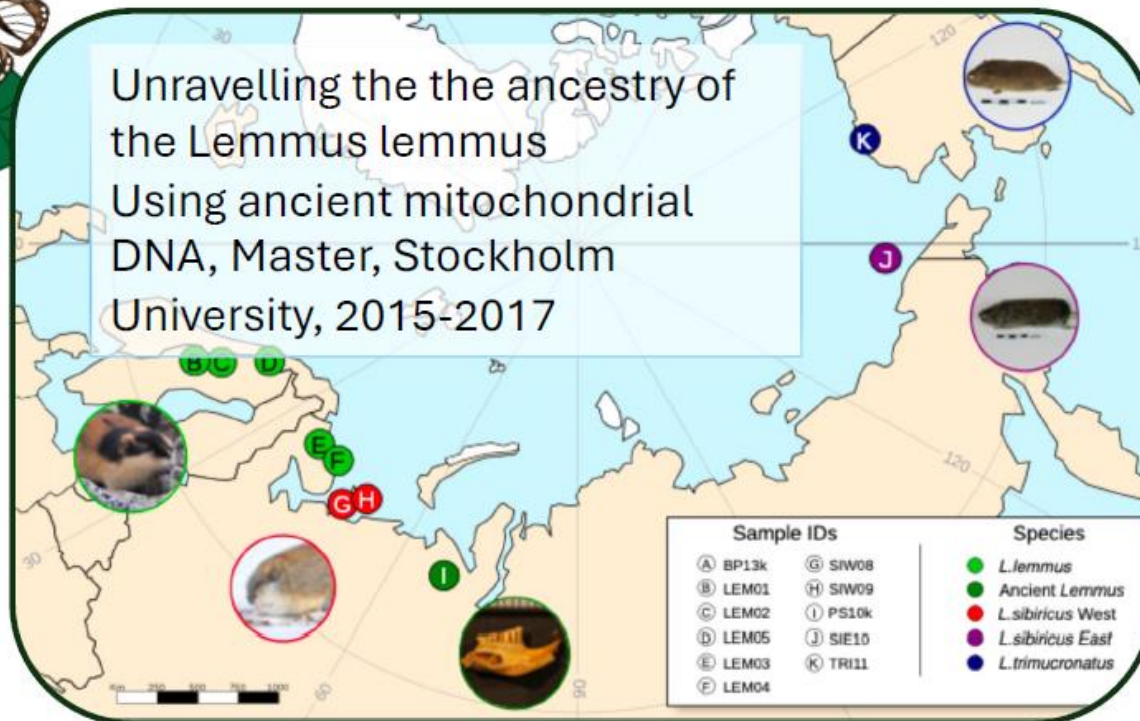
Karin Näsval, PhD

Postdoctoral fellow

Rapid speciation group, Tree of Life
Wellcome Sanger Institute, UK

Unravelling the the ancestry of
the Lemmus lemmus

Using ancient mitochondrial
DNA, Master, Stockholm
University, 2015-2017



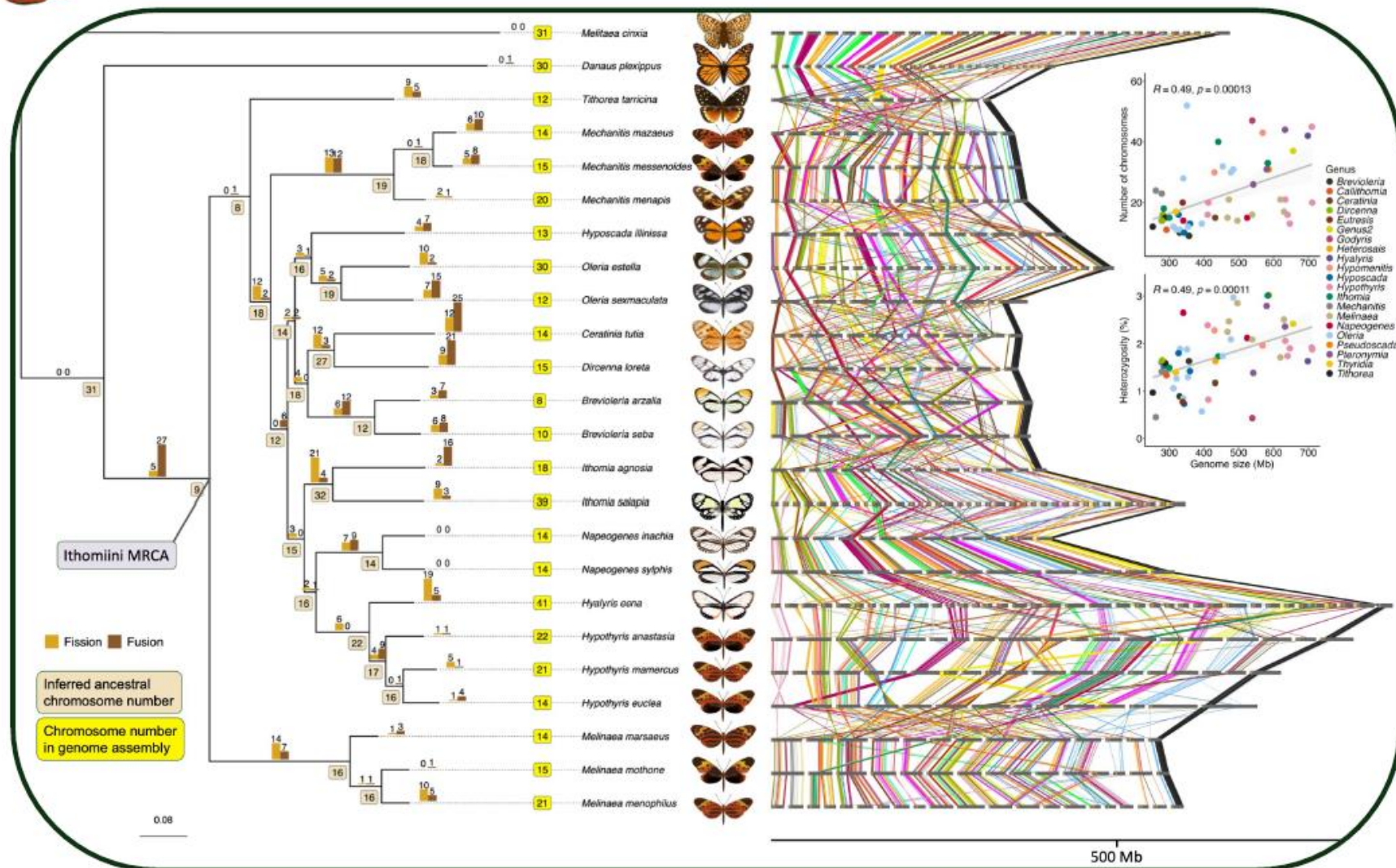
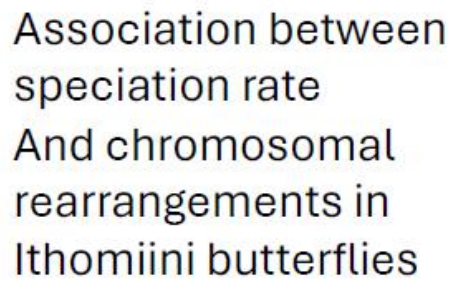
Doctoral student
2018-2023
EBC, Uppsala
University



Rates and patterns of genome
evolution in Lepidoptera

KARIN NÄSVALL





Fernando Seixas | My timeline

2010



Bachelor of Science in Biology



2013



Master of Science in Biodiversity, Genetics and Evolution
(Supervisors: José Melo-Ferreira & Paulo Célio Alves)



My timeline

2010 Bachelor of Science in Biology

U.PORTO

2013 Master of Science in Biodiversity, Genetics and Evolution
(Supervisors: José Melo-Ferreira & Paulo Célio Alves)

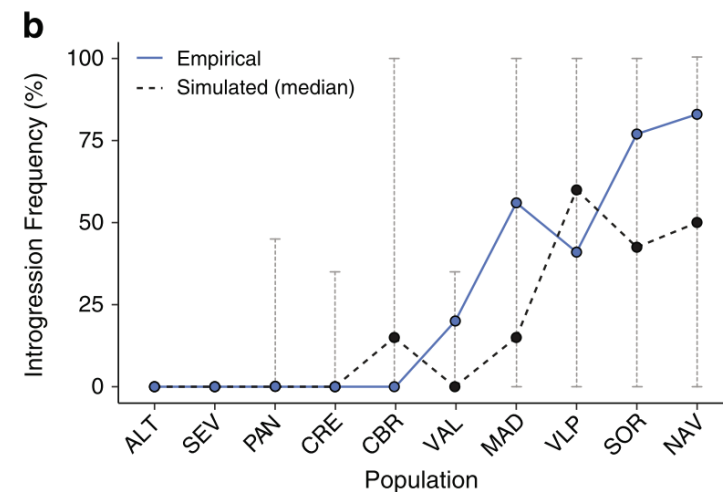
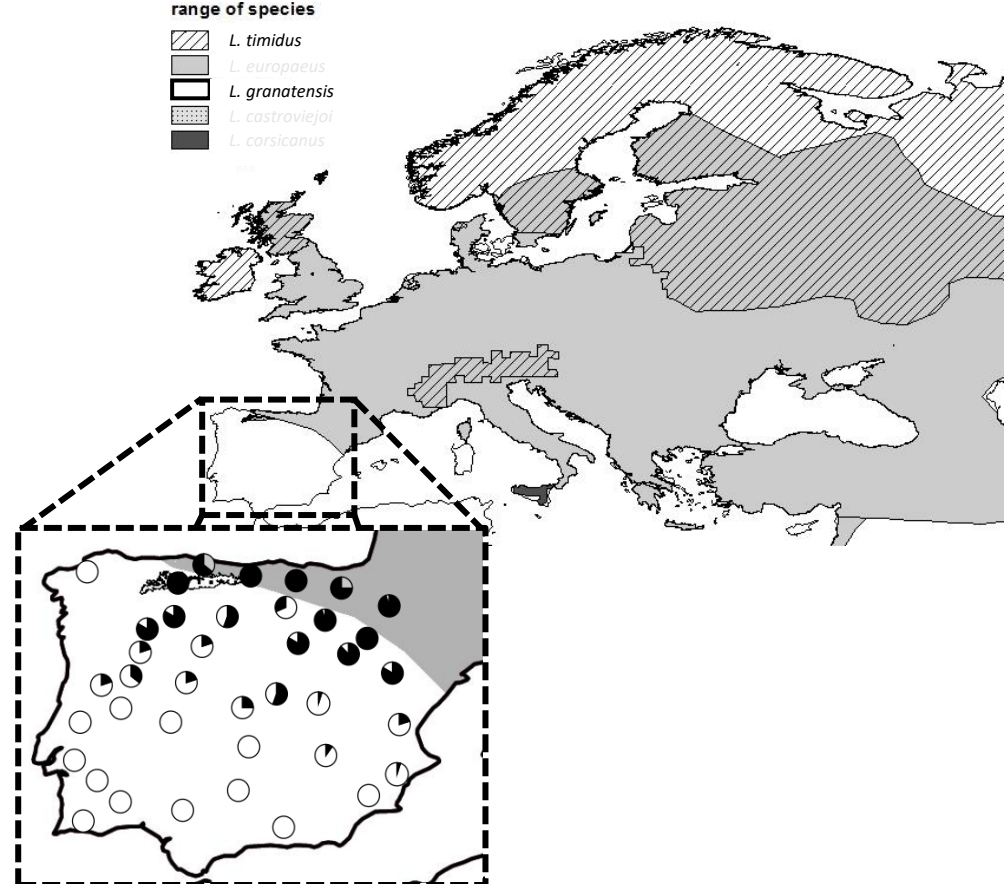


U.PORTO

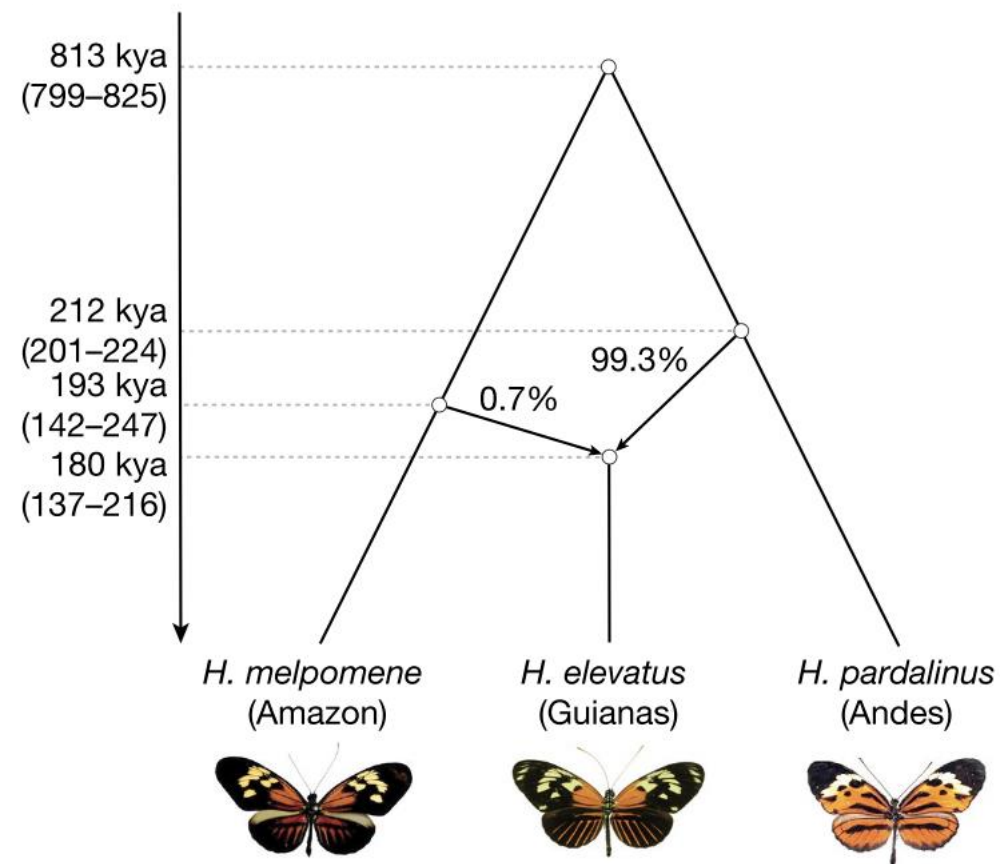
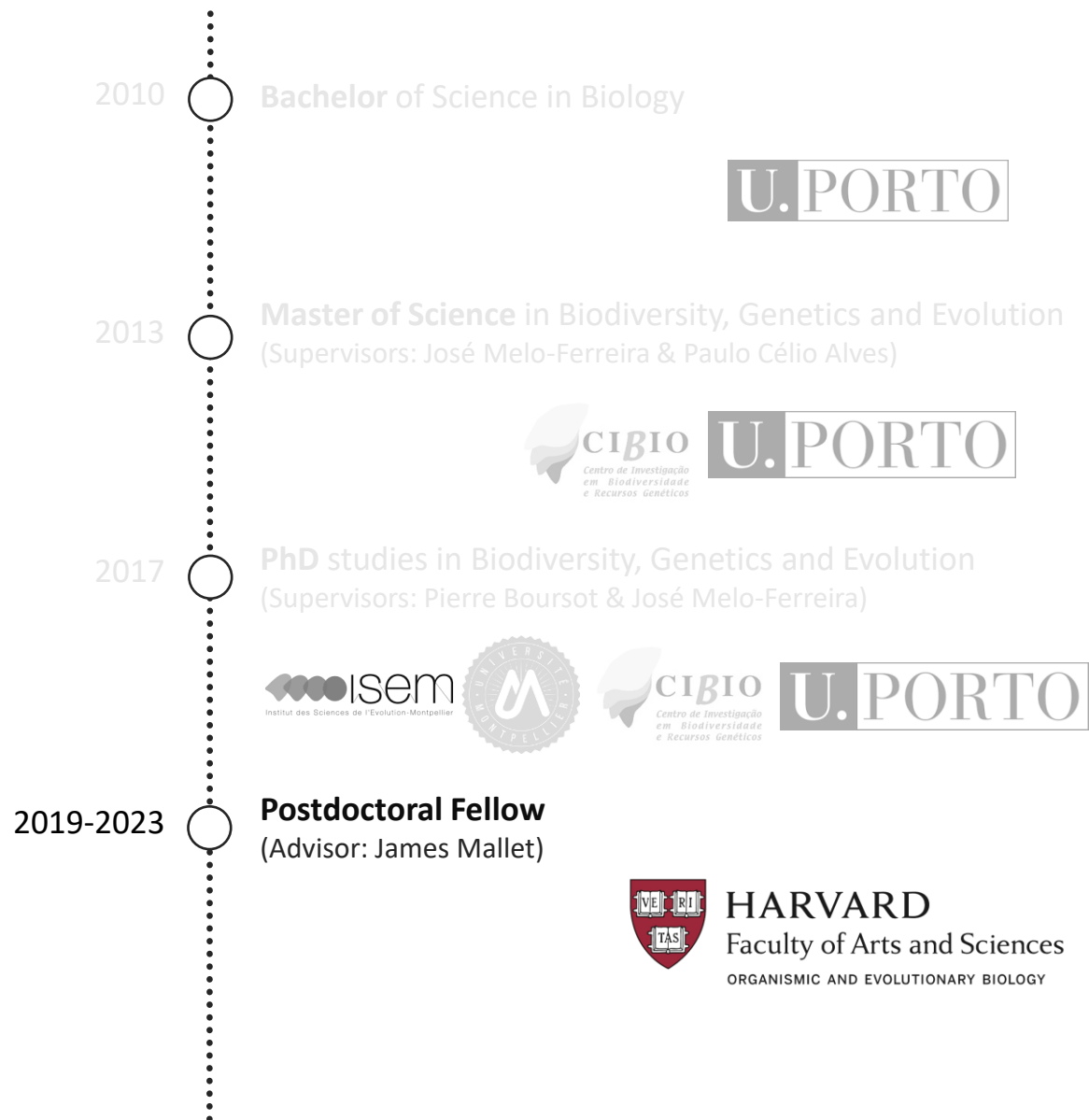
2017 PhD studies in Biodiversity, Genetics and Evolution
(Supervisors: Pierre Boursot & José Melo-Ferreira)



U.PORTO



My timeline



My timeline

2010

Bachelor of Science in Biology

U.PORTO

2013

Master of Science in Biodiversity, Genetics and Evolution
(Supervisors: José Melo-Ferreira & Paulo Célio Alves)



U.PORTO

2017

PhD studies in Biodiversity, Genetics and Evolution
(Supervisors: Pierre Boursot & José Melo-Ferreira)



U.PORTO

2019-2023

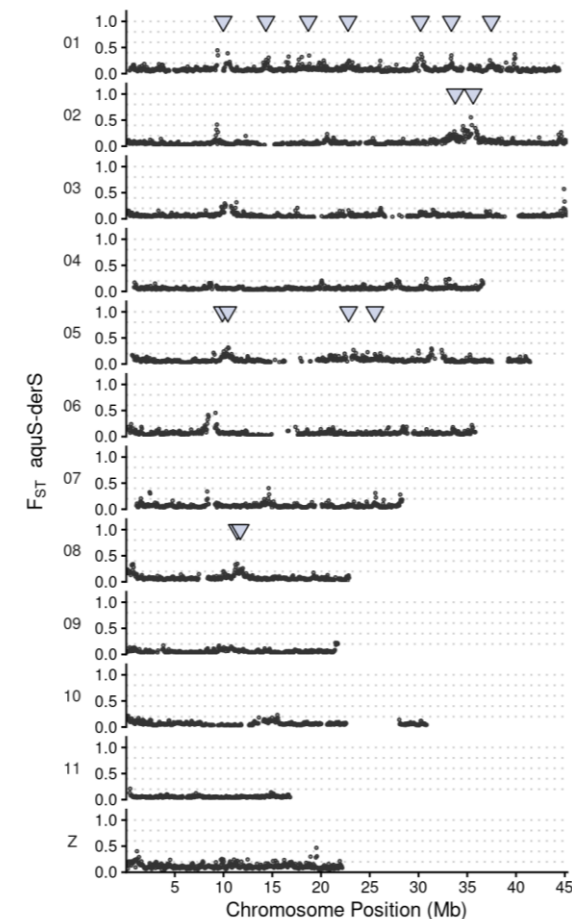
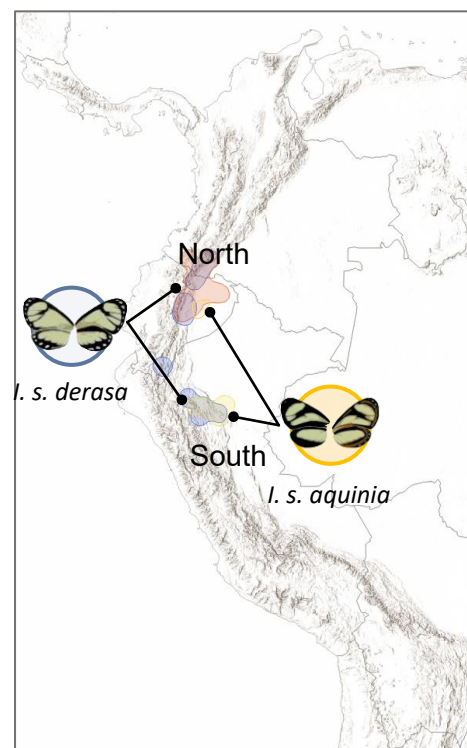
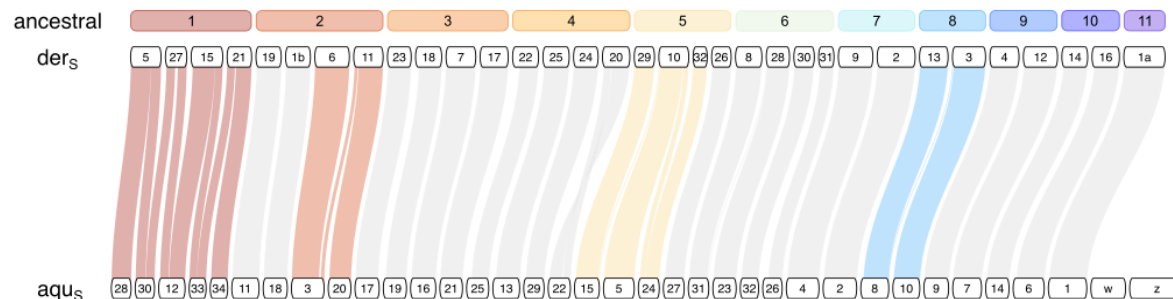
Postdoctoral Fellow
(Advisor: James Mallet)



HARVARD
Faculty of Arts and Sciences
ORGANISMIC AND EVOLUTIONARY BIOLOGY

2024-present

Postdoctoral Researcher
(Advisor: Joana Meier)



Ecosistemas estacional de Colombia



1

ESPECIACIÓN RÁPIDA

Aplicando genómica de poblaciones a la radiación en ecosistemas de alta montaña extrema.



2

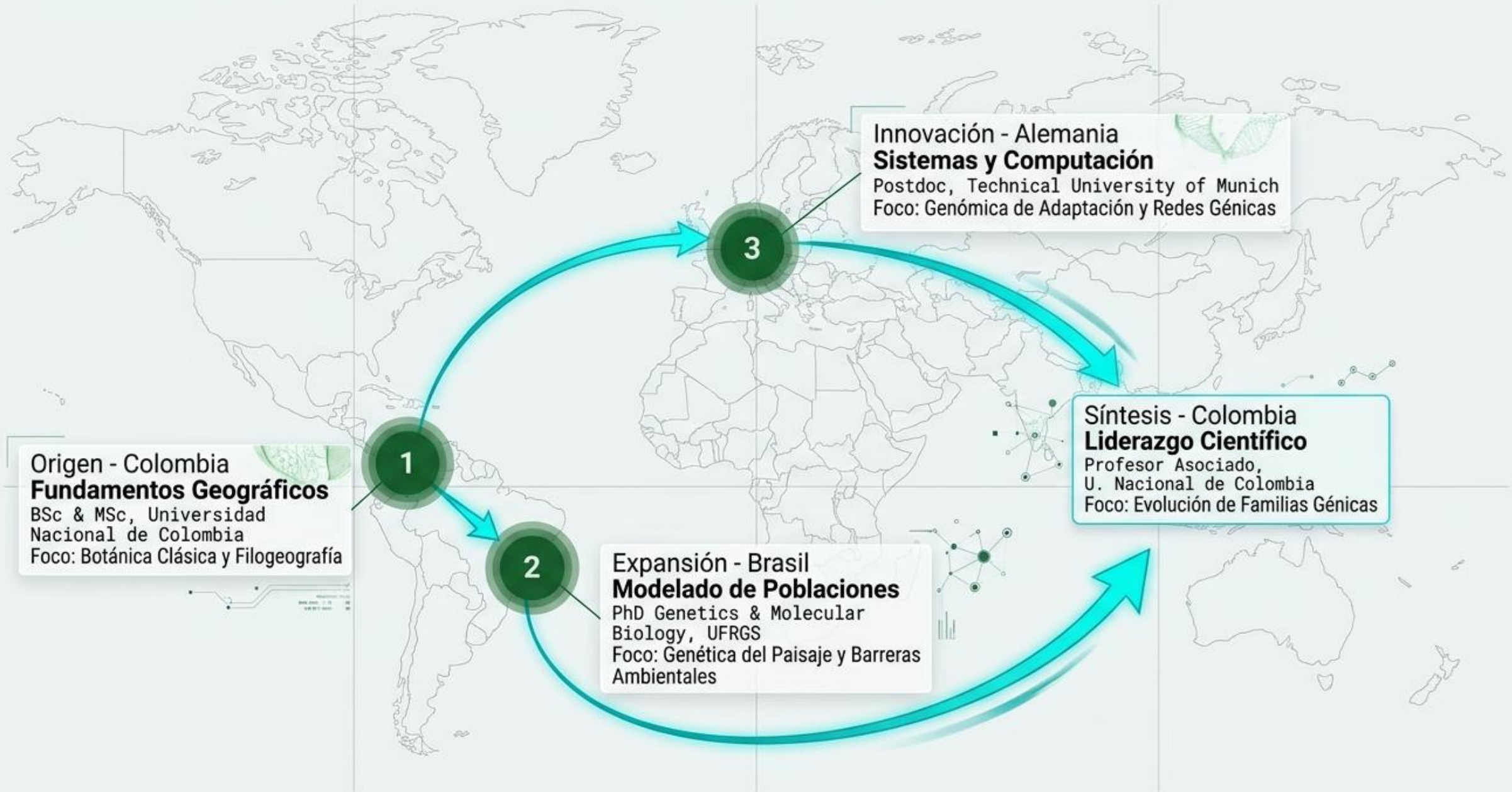
EVOLUCIÓN DE FAMILIAS GÉNICAS

Descubriendo redes de adaptación en árboles tropicales bajo estrés estacional.



Traer infraestructura analítica de clase mundial a uno de los puntos calientes de biodiversidad más complejos del planeta.

Trayectoria Académica: Una Evolución en Escala y Resolución



El Eje de Investigación: La Mecánica de la Adaptación

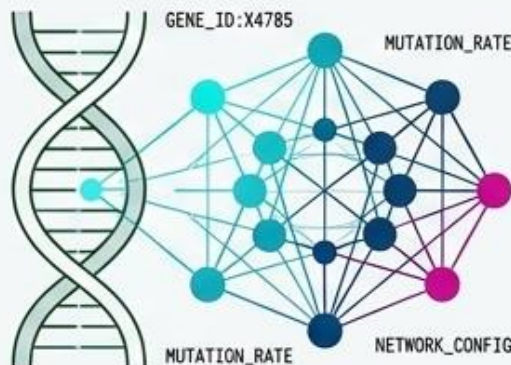
01. El Estrés Ambiental (Macro)



Cómo los gradientes climáticos y geográficos actúan como presiones selectivas.



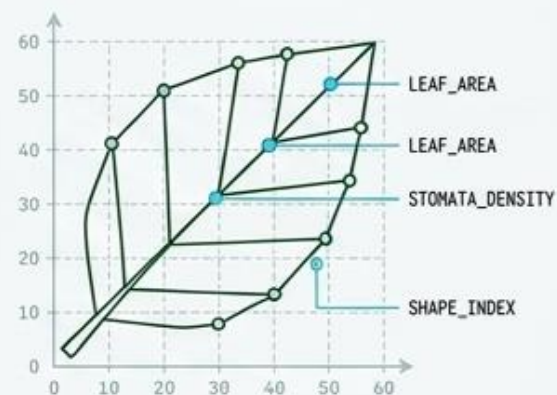
02. La Arquitectura Genómica (Micro)



Cómo las redes de genes mutan, se adaptan y se reconfiguran para sobrevivir.



03. La Expresión Fenotípica (Meso)



Cómo la morfología física de la planta cambia como respuesta medible.

Una búsqueda singular: descifrar el algoritmo exacto mediante el cual las plantas conquistan hábitats extremos.

La Transformación Metodológica: Fenotipado Computacional



1990s: Botánica Analógica.



2010s: Ensayo Físico.



2020s: Visión Computacional.

Introduction to Biodiversity Genomics

Using genomics to understand and preserve biodiversity from genetic diversity, populations, to species and ecosystems

Resolving the taxonomy

- Placing potentially new species
- Species delineation



Adaptation and speciation

- Identifying genomic regions involved in speciation
- Identifying genes underlying traits



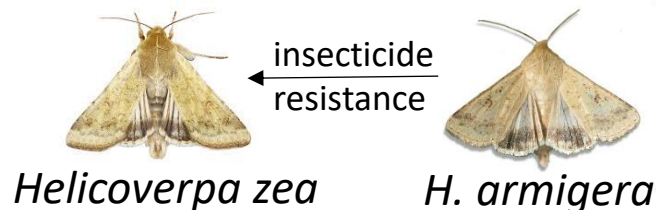
Informing conservation

- Assessing genetic diversity
- Assessing connectivity



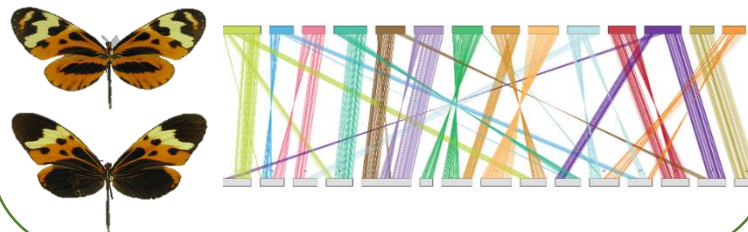
Gene flow

- Are populations/species hybridising or have in the past?
- Finding regions of adaptive introgression



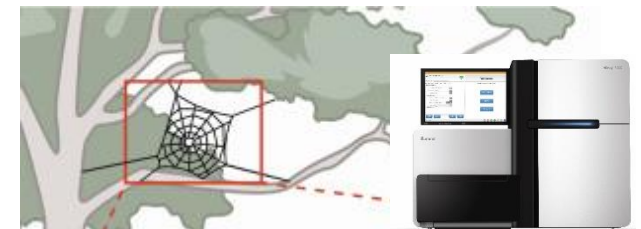
Genome evolution

- Gene expansions, e.g. olfactory
- Chromosomal rearrangements
- Genome size evolution (TEs, etc)

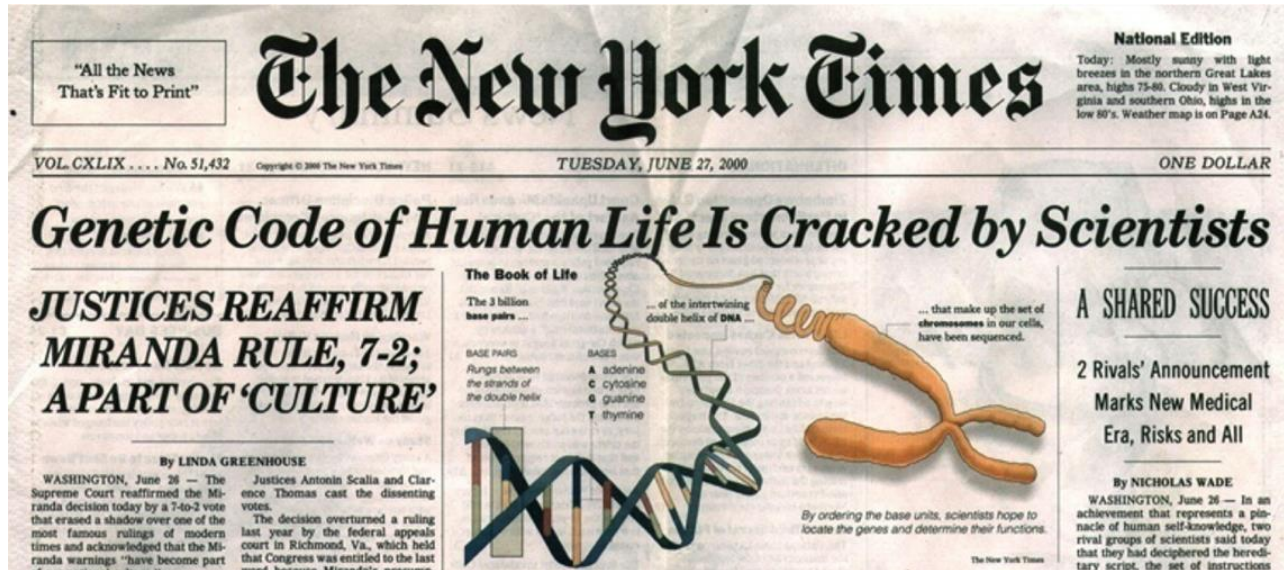


Which species occur here?

- Identifying biodiversity hotspots
- Monitoring effectiveness of conservation strategies

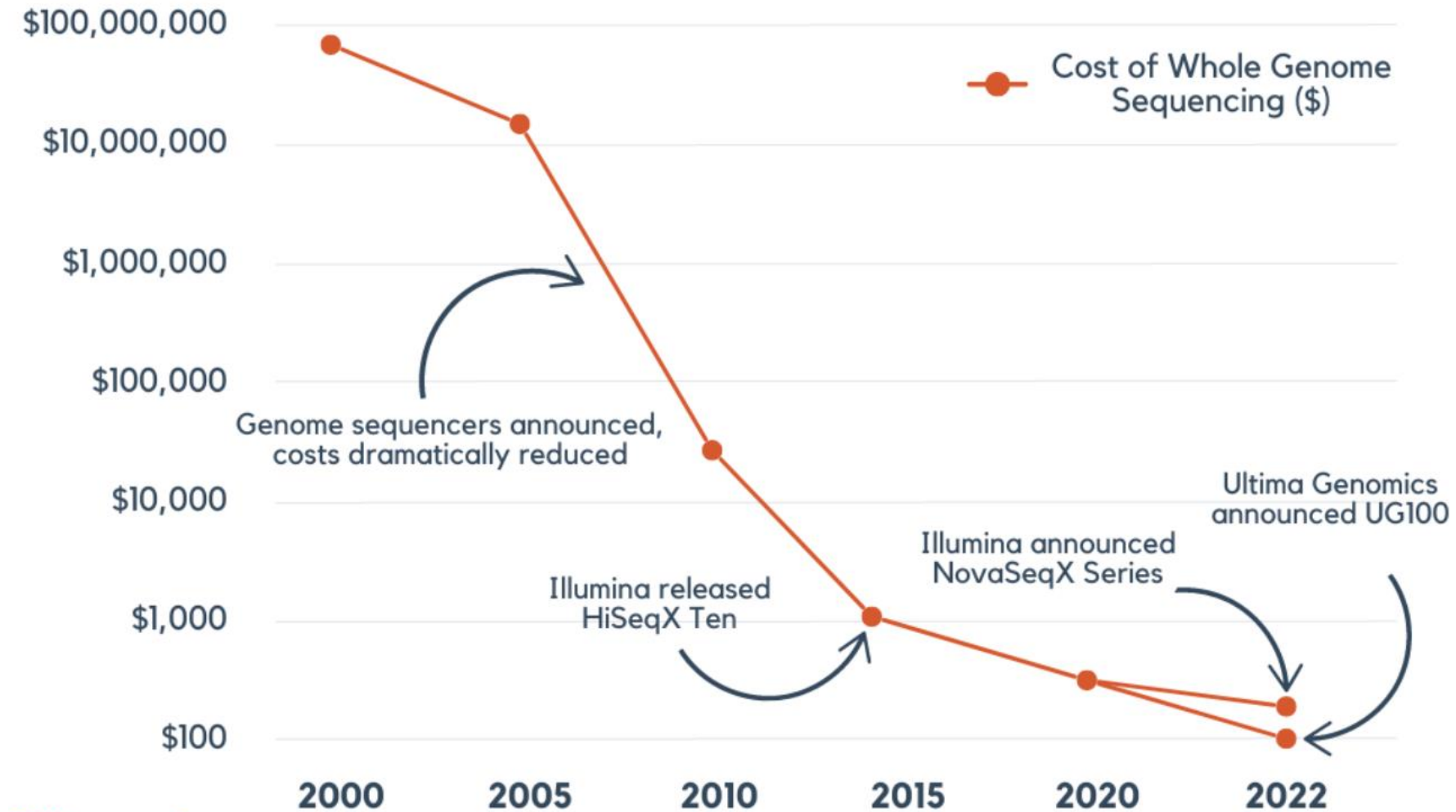


Human Genome Project



- Human genome project – started in 1990, completed in 2003
- Sequenced across ~20 institutions worldwide
- Cost an approximate \$5 billion US dollars

Sequencing costs are decreasing rapidly





CREATING A NEW FOUNDATION FOR BIOLOGY

Sequencing Life for the Future of Life



How do we make a reference genome?

Genome = set of all chromosomes



Break into many million
long (10-50 kbp) DNA fragments



Long-read sequencing
machine (e.g. PacBio)

more expensive than
short-read sequencing



**Millions of reads,
10-20 kbp long**

GATGCTGAGTA
ATAGTGTGGAT
GTGTGGATGTG
TGCTGAGTTCG
TGGATGCTGAT
CTGAGTTCTCG

Reference genome

ATAGTGTGGATGCTGAGTTCGT

ATAGTGTGGATG
GTGTGGATGCT
TGGATGCTGAGT
GATGCTGAGTTC
TGCTGAGTTCG
CTGAGTTCGT

puzzling them together
(aligning them to
each other)

Draft genome

Contig 1

ATAGTGTGGATGCTGAGTTCGT

ATAGTGTGGATG
GTGTGGATGCT
TGGATGCTGAGT
GATGCTGAGTTC
TGCTGAGTTCG
CTGAGTTCGT

Contig 2

TTCAGGGGCATAAAAGTAACGTA

TTCAGGGGCAT
AGGGGCATAAAA
GGCATAAAAGTAACG
CATAAAAGTAACGTA

Contig 3

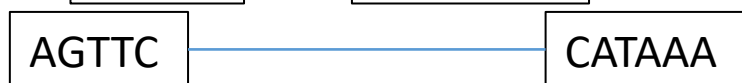
GACACCCGCATATTAGTAAC

GACACCCGCAT
ACCCGCATATTA
CGCATATTAGTA
CATATTAGTAAC

Adding Hi-C data

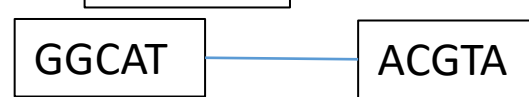
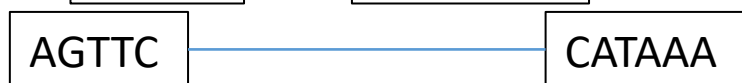
Contig 1

ATAGTGTGGATGCTGAGTTCGT



Contig 2

TTCAGGGGCATAAAAGTAACGTA



Contig 3

GACACCCGCATATTAGTAAC



Contig 4

ATCGTA...



Scaffold 2

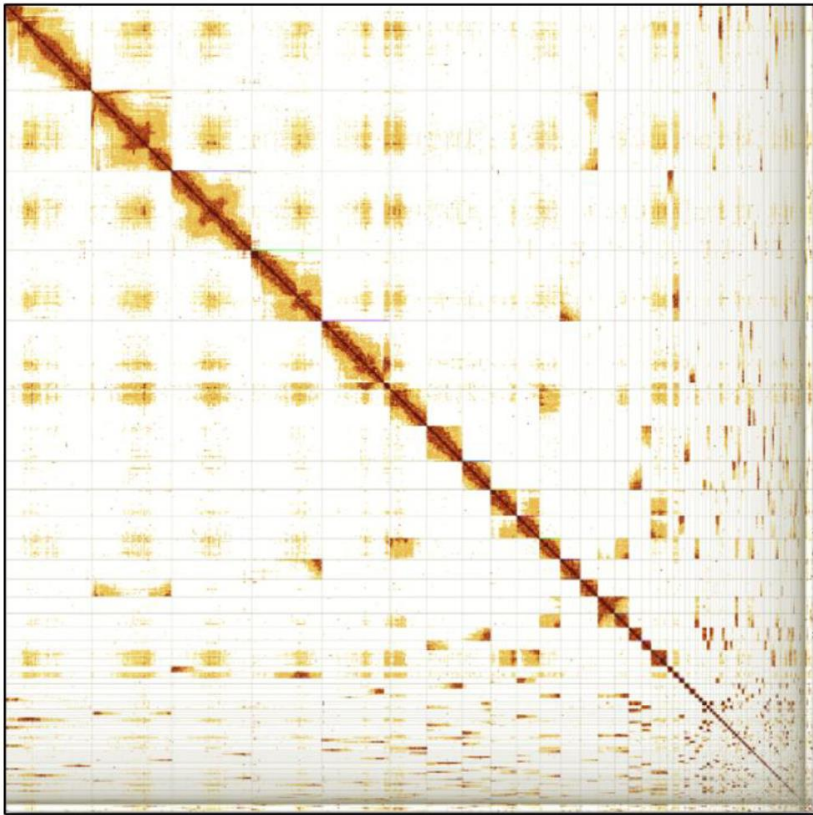
ATAGTGTGGATGCTGAGTTCGT **NNN** TTCAGGGGCATAAAAGTAACGTA

Scaffold 2

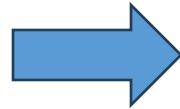
GACACCCGCATATTAGTAAC **NNN** ATCGTA...

Manual curation

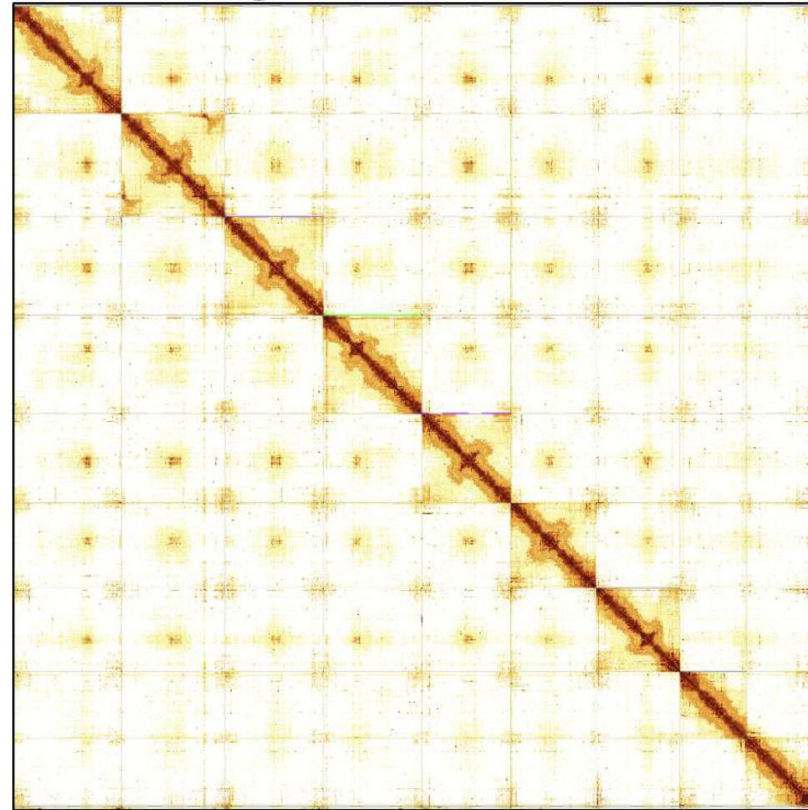
n = 230
N50 = 33.1Mb



Pre curation assembly



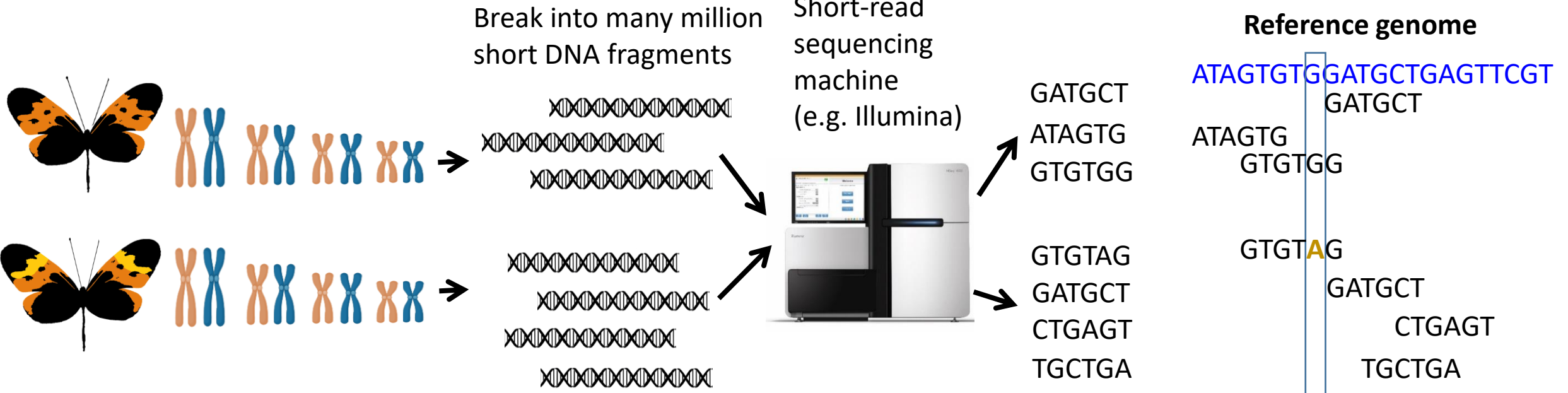
n = 62
N50 = 87.1Mb
99.85% of genome in 9 Chromosomes



Post curation

Whole genome resequencing with short reads

Genome = set of all chromosomes

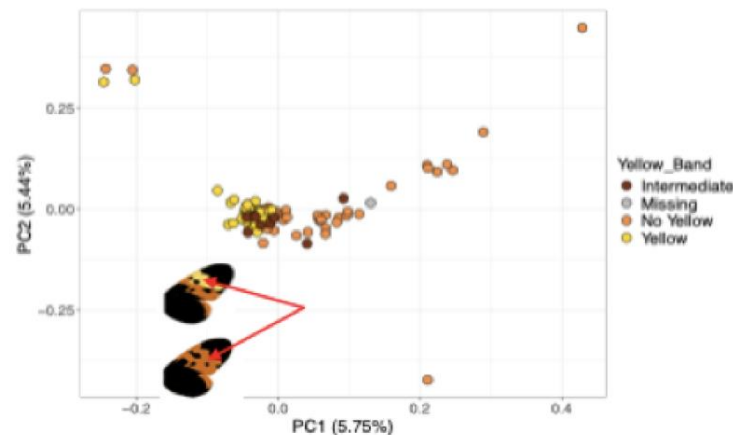




Do these two butterflies belong to different species?



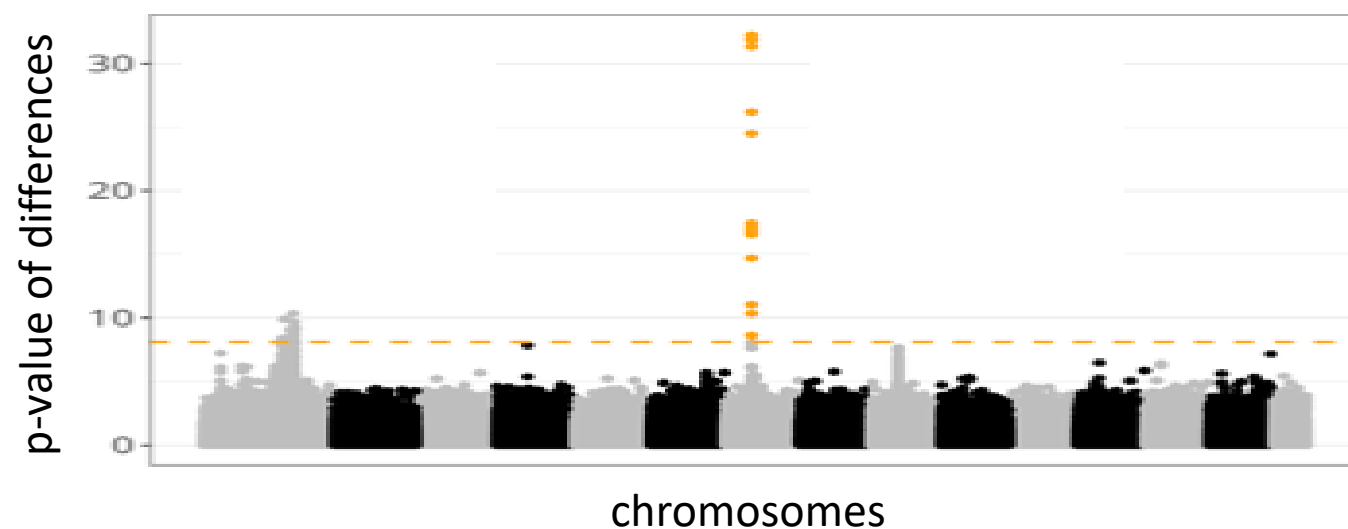
Eva van der Heijden



CRISPR butterflies with *ivory* knocked-out

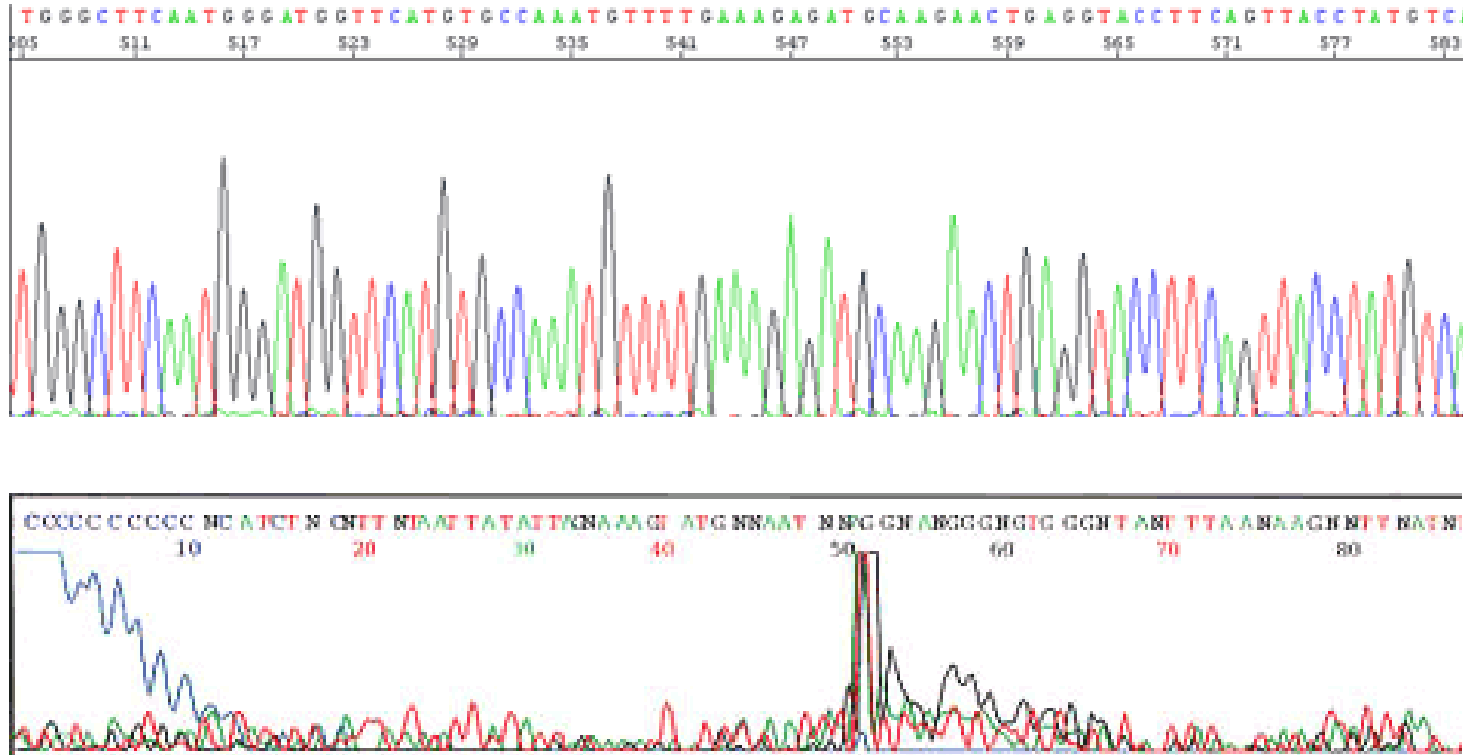


They differ at only one region in the genome



Introduction to high-throughput or next-generation sequencing

Sanger Sequencing (since 1980s)



- Possible to manually check each sequence and resequence failed sequences
- Requires primer sequences and has very low throughput (expensive per bp)

Two main types of high-throughput sequencing

- **Short-read sequencing**

- Reads are typically 150 bp long
- Cheaper than long-read sequencing
- E.g. Illumina, soon probably also Ultima Genomics

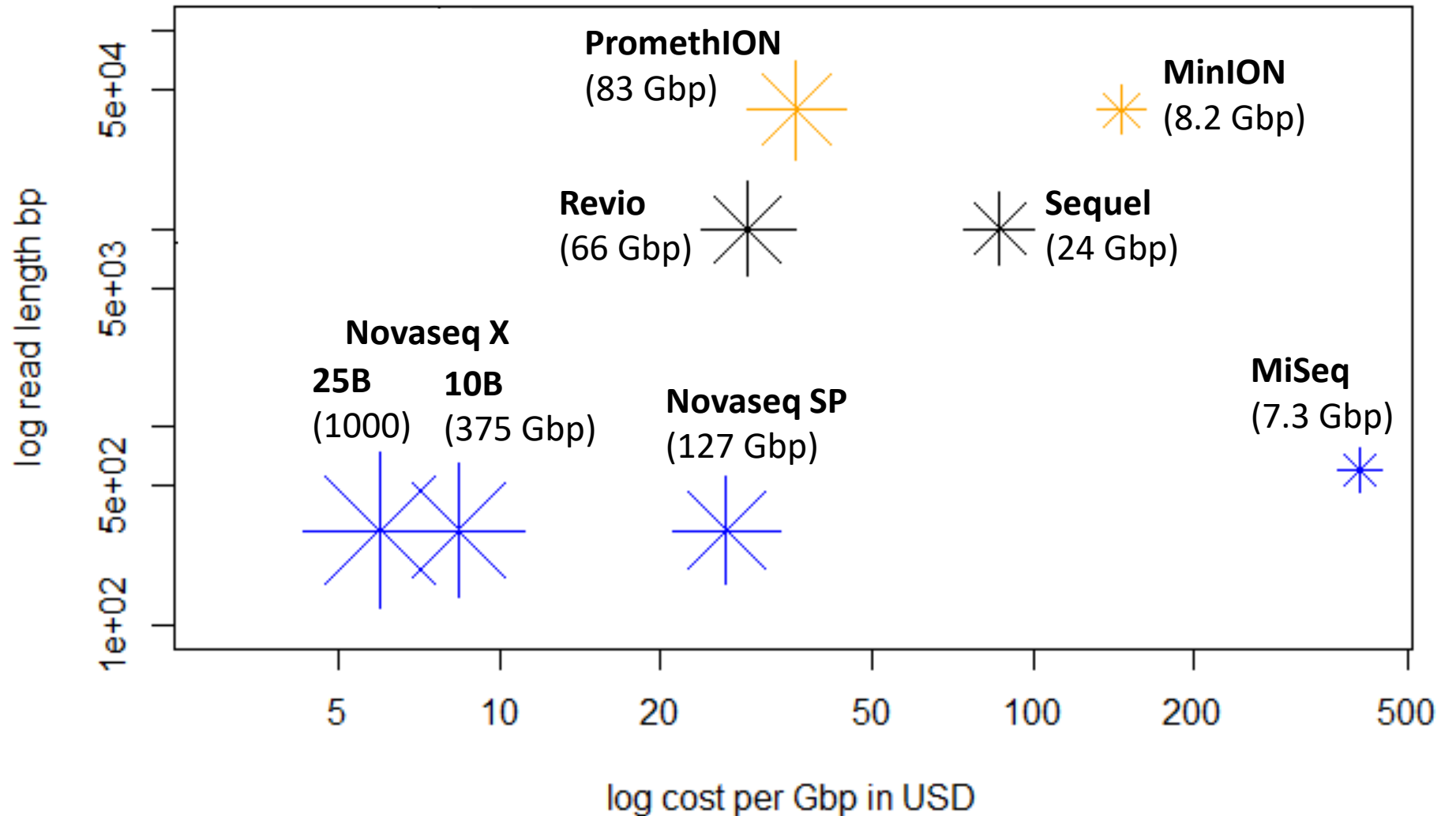
- **Long-read sequencing**

- Reads are typically >10 kb long (PacBio: 15-20 kbp, Nanopore: 10-100 kbp)
- More expensive than short-read technologies
- Required for making a reference genome
- E.g. PacBio or Nanopore

Read length versus per Gbp sequencing costs for different sequencing machines (note the axes are in logarithmic scale)

* ONT
* PacBio
* Illumina

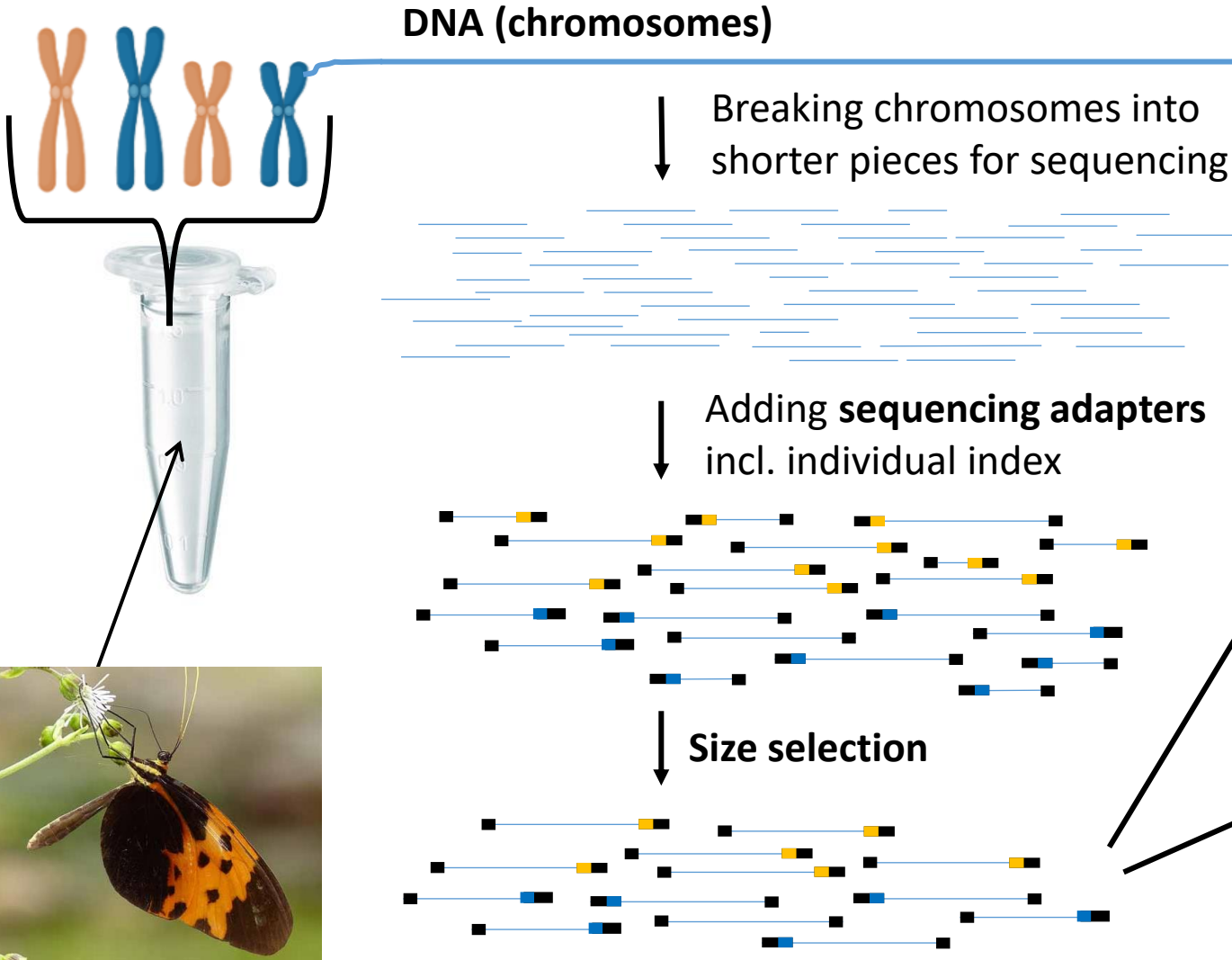
Star size shows the total throughput per lane, also given in parentheses ()



Whole-genome sequencing

Genome

= complete set of chromosomes



Long-read sequencing (PacBio or Nanopore/ONT) paired-end sequencing



10-50 kbp

Short-read sequencing (e.g. Illumina) paired-end sequencing



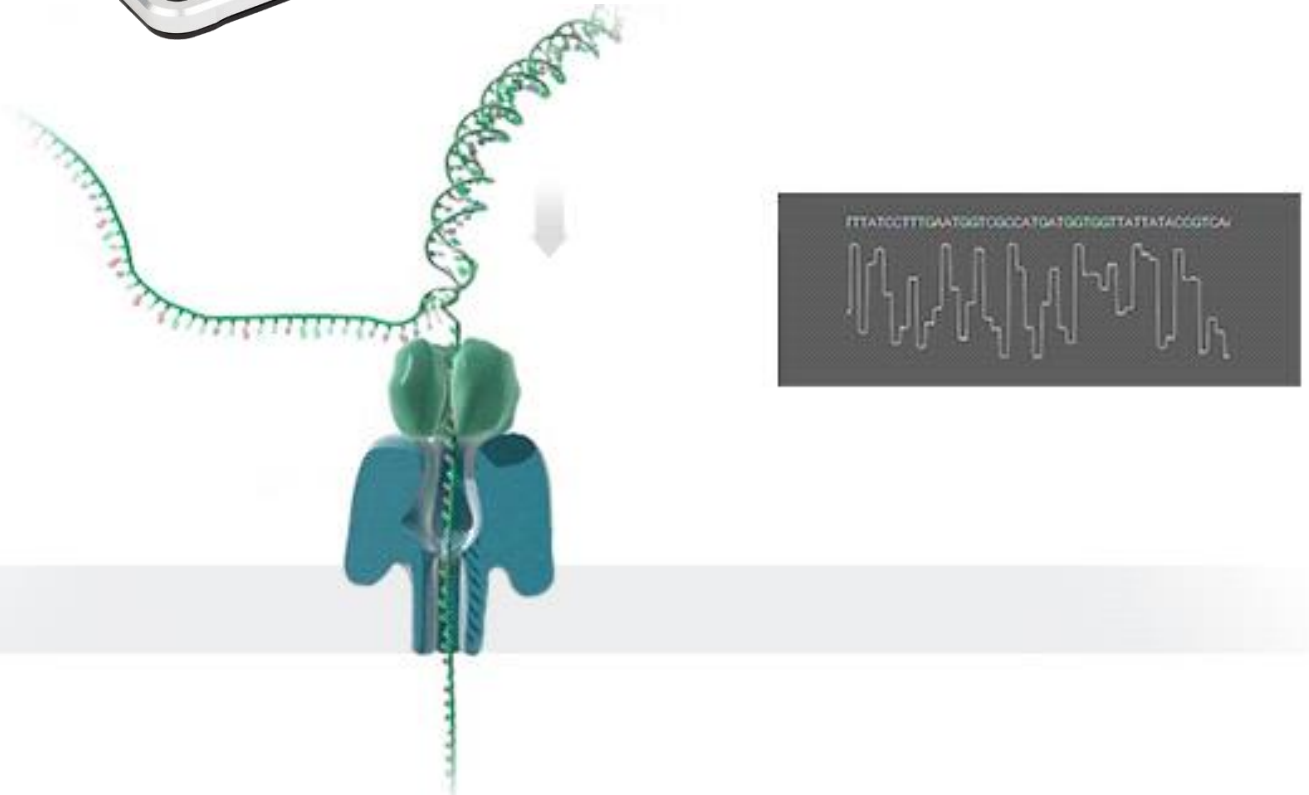
350-500 bp

150 bp 150 bp

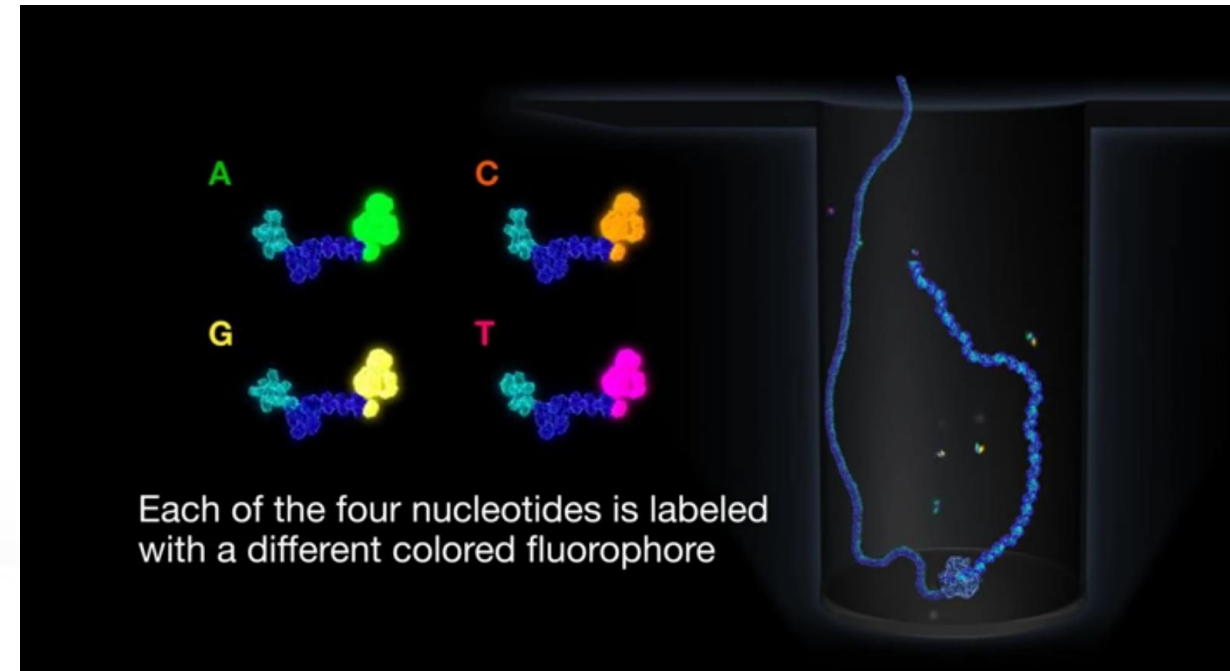
Long read sequencing technologies



Nanopore



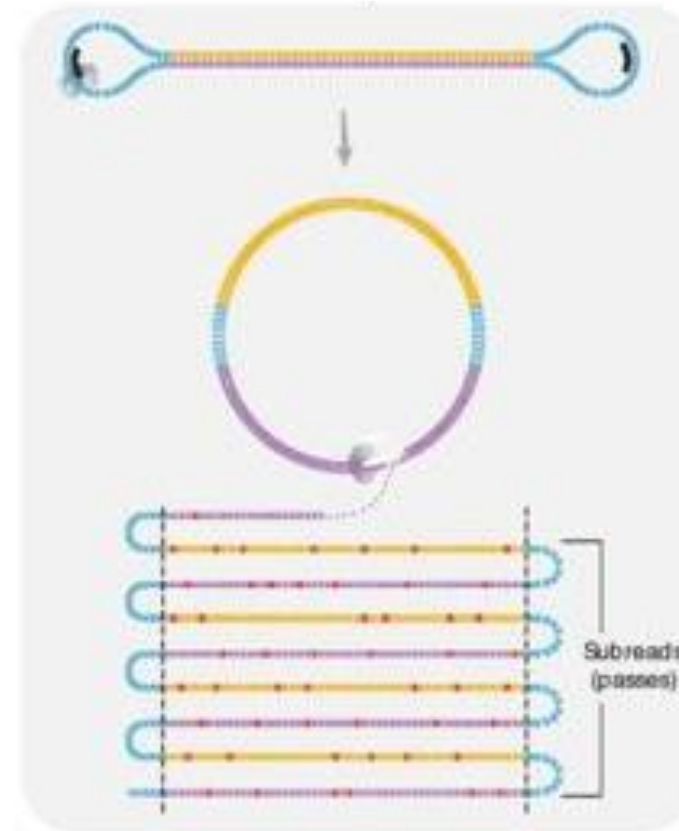
PacBio



PacBio HiFi reads (99.95% accuracy)

Each DNA-fragment is sequenced many times to get a high-quality consensus (=summary) read

Multi-pass sequencing
on Sequel II System



HiFi Read Base Calling

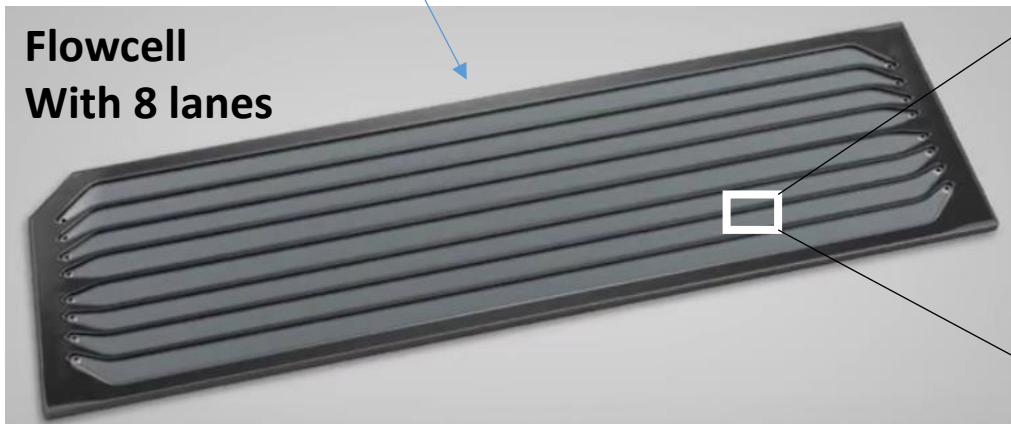
HiFi READ
Read Lengths up to 25,000 bp
Average Read Accuracy $\geq 99.5\%$



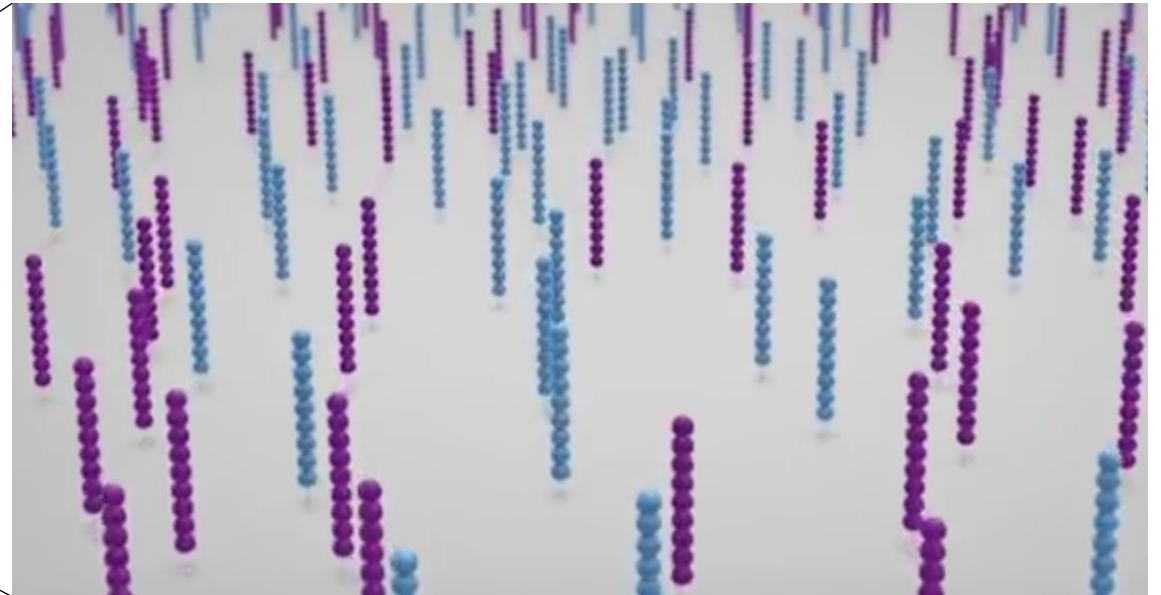
Illumina flowcell

DNA fragments with Illumina adapters

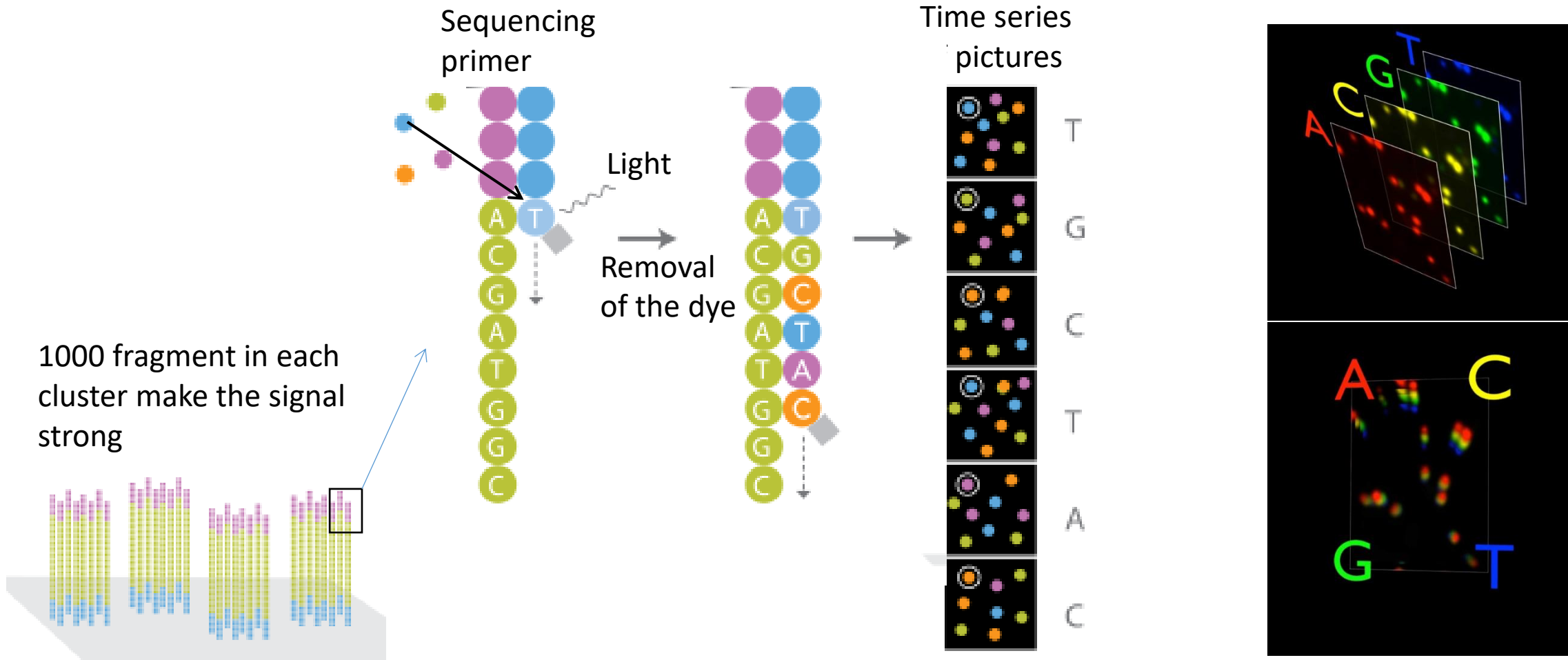
Flowcell
With 8 lanes



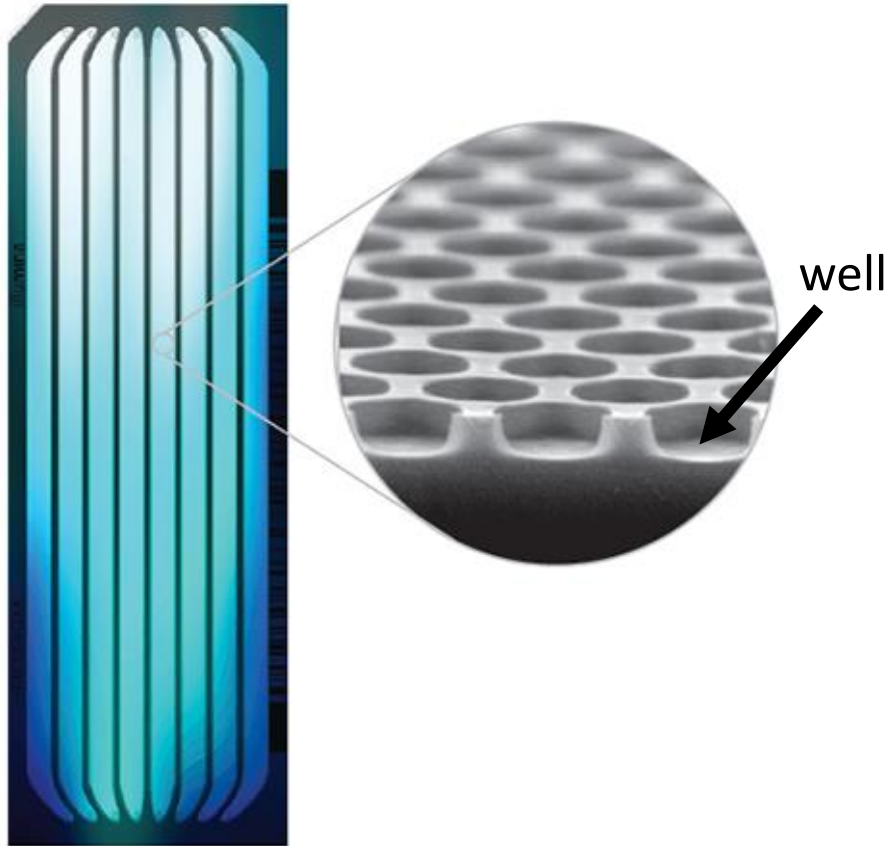
Each lane contains a dense lawn of Illumina primers



Short-read sequencing with Illumina



**Newer Illumina machines use wells and only 2 colours
(e.g. Novaseq, Nextseq, MiniSeq. This makes it faster and cheaper)**

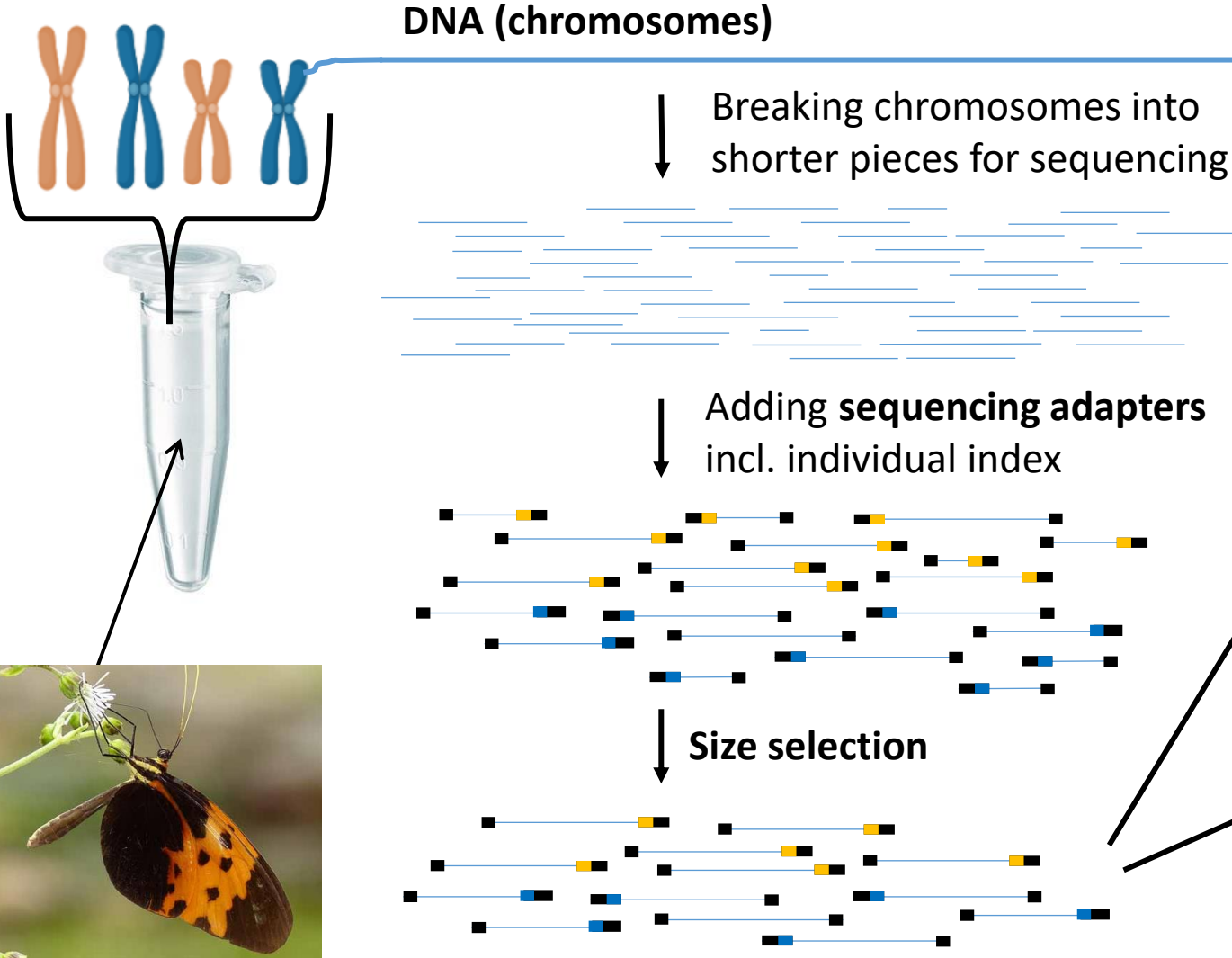


2-Channel Chemistry				
	 A	G	 T	 C
Image 1				
Image 2				
Result	A	G	T	C

Whole-genome sequencing

Genome

= complete set of chromosomes



Long-read sequencing (PacBio or Nanopore/ONT) paired-end sequencing



10-50 kbp

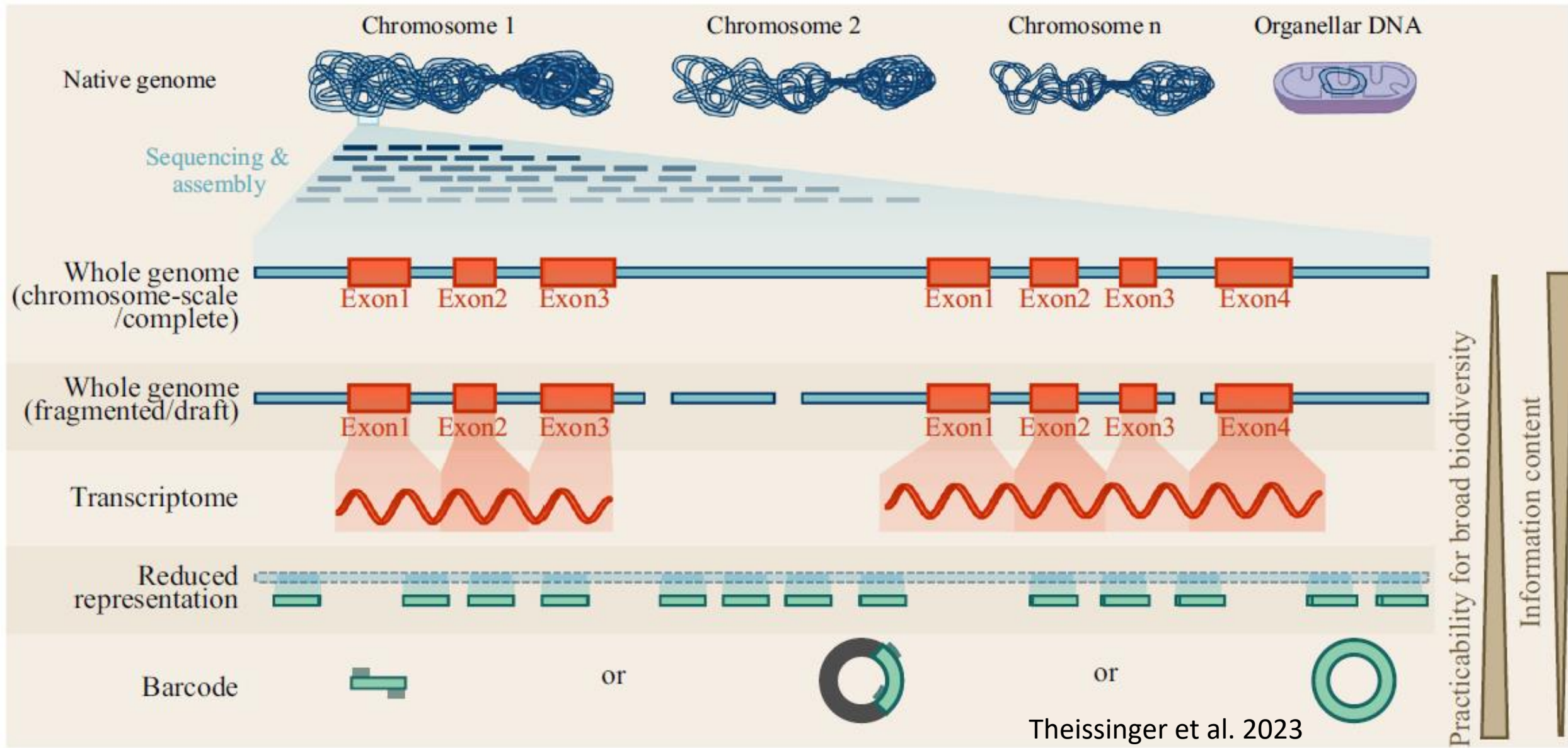
Short-read sequencing (e.g. Illumina) paired-end sequencing



350-500 bp

150 bp 150 bp

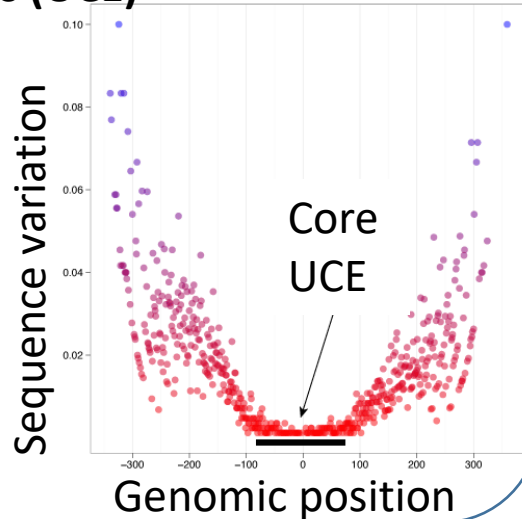
Sequencing approaches for biodiversity genomics



Reduced-representation techniques (only parts of the genome sequenced)

Ultra-conserved elements (UCE)

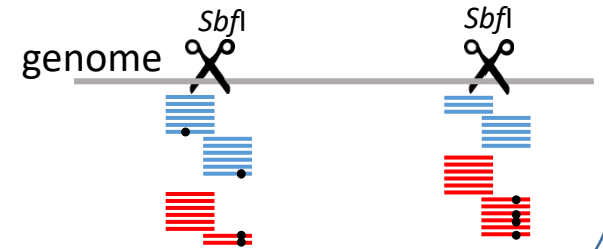
- Sequence capture with baits based on genomic regions that are conserved across many species
- Works with highly divergent species



Restriction Associated Sequencing (RAD)

(similar methods: GBS, ddRAD)

- does not require primers/baits or reference genome
- individuals need to be from the same or closely related species
- Information from thousands of loci distributed across the genome

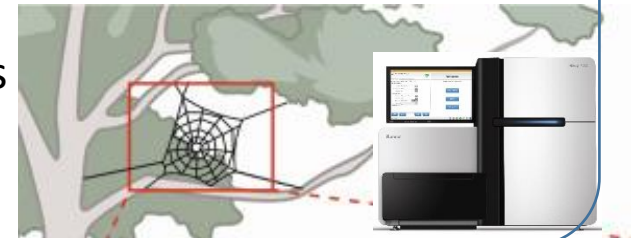


Targeted or amplicon sequencing, e.g. barcoding

- Sequencing one or few genes
- requires primers
- e.g. CO1 (mitochondrial barcoding region), advantage: large database (BOLD) available to compare to for species identification

Environmental DNA (eDNA)

- Mostly CO1 sequencing from soil, water, air (spider webs)
- Identifying local species
- Studying species richness



Trade-offs: Splitting reads (i.e. costs) among:

Total data gets divided by:

- Number of sites to sequence
 - Depends on genome size and sequencing strategy, e.g. RAD versus whole-genome
- Sequencing depth (e.g. sequencing at 10x depth of coverage)
- Number of specimens to sequence
- Example: 1 Novaseq X 10 B lane
~2.5 billion paired-end reads of 150 bp each -> 375 Gbp data
 - 100 whole-genomes of a species with 0.375 Gbp genome size at 10x coverage
 - 19 whole-genomes of a species with 1 Gbp genome size at 20x coverage
 - 375 individuals sequenced with a RAD sequencing approach resulting in 50 Mbp at a sequencing depth of 20x

**Now let's get started with
handling genomic data!**

**First, a brief overview of what
we will do now**

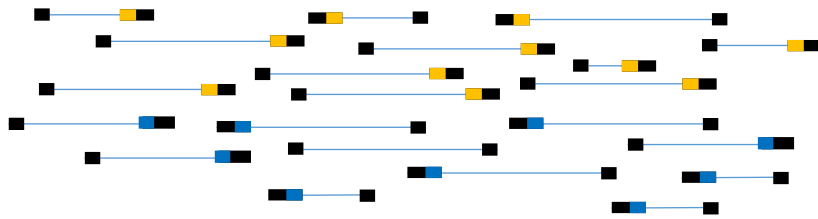
Whole-genome short-read sequencing

DNA

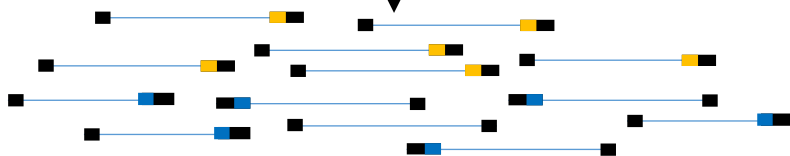
↓ Random shearing



↓ Illumina adapter ligation
incl. individual index



↓ Size selection



paired-end sequencing

Up to 20 billion read pairs

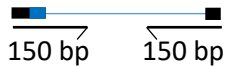


1. Quality check and trimming raw reads

paired-end Illumina sequencing



Up to 20 billion
read pairs



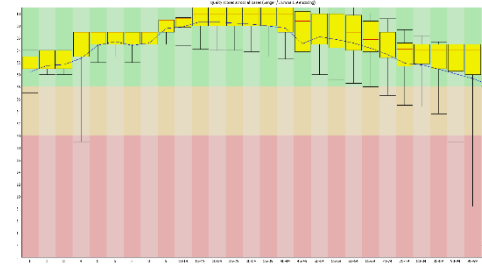
Forward
reads



Reverse
reads

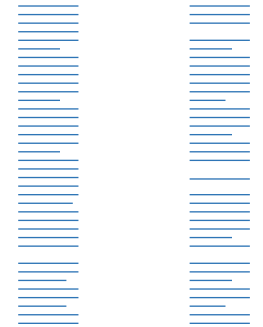


Fastqc quality plots



fastp

Trimmed and filtered
reads of good quality



Fastq file of reads

```
@HWUSI-EAS611:34:6669YAAXX:1:1:5069:1159 1:N:0:  
TCGATAATACCGTTTTTTTCCGTTTGATGTTGATACCAT  
+  
IIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHIIHHII
```

From samples to a vcf file

